

회로이론과 계측법(이론) (BE205-02)

학기	2026년 1학기
담당교수	송민영
이수구분	교선�필수
학점	2.0
강의시간	화17:00-19:00(E7 - 233)
강의실	E3 610 (교수실)
소속	기초학부

교과개요

This course introduces the fundamental principles and analytical techniques of electrical circuit theory. Topics include basic electrical quantities and circuit elements, core circuit analysis methods such as Kirchhoff's laws, transient and AC response analysis, Laplace transform-based circuit analysis, frequency response, basic system concepts, and AC power analysis. Through this course, students will develop a systematic understanding of the fundamental principles and analytical methods of electrical circuits.

교과목표

By the end of this course, students will be able to:

- Explain fundamental electrical quantities and circuit elements and describe their physical behavior in electrical networks.
- Analyze linear circuit networks using core circuit analysis
- Understand transient and AC behavior of circuit networks
- Understand and analyze frequency response characteristics and basic system concepts of linear circuits (e. g., filters, amplifiers).
- Calculate and interpret AC power quantities, including average power, reactive power, and power factor.

수업방법

Lectures and homework

학습윤리

No plagiarism

교과정책(방침)

It is recommended to take this course together with Circuit Theory and Measurements (Lab), which will be open in the next semester, but it is not mandatory.